

Tempus Oncology Product Knowledge Base

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Pages fetched: /oncology/, /oncology/genomic-profiling/xt-xr/, /oncology/genomic-profiling/xt/, /oncology/genomic-profiling/xt-heme/, /oncology/algorithmic-tests/, /oncology/hub/, /oncology/tempus-one/, /oncology/clinical-trial-matching/, /oncology/ehr-integration/, /pathology/

Company Overview

Tempus is a healthcare technology company advancing the use of AI in precision oncology. By integrating comprehensive molecular and clinical data, Tempus delivers actionable insights that help oncologists make more informed decisions throughout the patient journey.

Tempus offers a robust portfolio of tests — from germline and somatic testing to MRD and treatment response monitoring — providing a complete patient view to support clinical decisions.

Scale:

- 6,500+ oncologists rely on Tempus
 - 40M+ records across Tempus
 - 4,500+ connected healthcare institutions
 - 40+ operational countries
 - Partners with more than 50% of all academic medical centers in the United States
 - 2,000+ publications in leading scientific journals
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Genomic Profiling Tests

1. xT CDx — FDA-Approved Solid Tumor DNA Sequencing

What it is: An FDA-approved 648-gene tissue-based NGS test for molecular profiling of all malignant solid tumors. Intended as a companion diagnostic (CDx) to identify patients who may benefit from treatment with targeted therapies.

Sample type: Formalin-Fixed Paraffin Embedded (FFPE) tumor tissue + matched normal blood or saliva

What it detects:

- Substitutions (SNVs and MNVs) and insertion/deletion alterations (INDELs) in 648 genes
- Microsatellite instability (MSI) status
- Tumor Mutational Burden (TMB)
- HLA Class I genotyping
- Copy number variations (CNVs)
- Rearrangements in 22 genes
- Potential germline findings
- Companion diagnostic (CDx) claims for colorectal cancer (CRC) patients

Key clinical evidence:

- 43.4% of patients matched to a targeted therapy when DNA seq, RNA seq, and immune biomarker assessment were combined, compared to 29.6% with DNA seq alone

- 28% reduction in somatic false-positive calls, improving accuracy compared to tumor-only analysis
- ~7% of cancer patients found to harbor pathogenic or likely pathogenic germline variants across multiple tumor types
- 96% of patients matched to a clinical trial when clinical data was combined with Tempus NGS

Ordering flexibility:

- Available as a standalone test or in addition to xR RNA seq and xF Liquid Biopsy
- Auto-conversion from xT CDx to xF Liquid Biopsy in the event of insufficient tissue
- Available for tumor-only testing including hematologic malignancies
- Orderable via Tempus Hub, paper requisition, or directly from your EHR

Performance Specifications (Chicago Lab):

Variant Type	Positive Percent Agreement (PPA)	Negative Percent Agreement (NPA)
SNVs	99.2%	100.0%
MNV	94.7%	100.0%
Insertion	96.7%	100.0%
Deletion	100%	100.0%
MSI	94.0%	98.0%

Add-on tests available with xT CDx:

- IHC: PD-L1 (22C3, 28-8, SP142, SP263), MMR (MLH1, MSH2, MSH6, PMS2), HER2, FOLR1 (FR α), CLDN18, c-MET
- Neuro-oncology: 1p/19q Co-Deletion, MGMT Promoter Methylation
- Algorithmic: IPS, HRD, Tumor Origin (TO), DPYD, UGT1A1, PurlST■

2. xR — Whole Transcriptome RNA Sequencing

What it is: A whole transcriptome RNA sequencing panel for solid tumors and hematologic malignancies.

What it detects:

- Clinically relevant fusions across more than 100 targeted genes, including ALK, RET, ROS1, NTRK1/2/3, FGFR1/2/3, NRG1, BRAF, EWSR1, ESR1-CCDC170, MYB-NFIB, PML-RARA, BCR-ABL
- Altered splicing events such as MET exon 14 and EGFRvIII

Key clinical evidence:

- 21% more patients with driver fusions eligible for FDA-approved targeted therapies identified with DNA + RNA sequencing compared to DNA sequencing alone
- 29% more patients identified with a unique clinically actionable fusion when RNA seq incorporated, compared to DNA seq alone
- RNA NGS increases ALK fusion detection by 18% over DNA alone for patients with advanced NSCLC

Ordering flexibility:

- Can be ordered standalone or in combination with xT CDx
- Add-ons available: IHC, FISH, MGMT Promoter Methylation analysis

Performance Specifications (Chicago Lab):

Variant Type	PPA	NPA
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Rearrangements/Fusions (targeted)	100.0%	99%
Rearrangements/Fusions (untargeted)	97.0%	99%
Altered Splicing (MET Exon 14)	100.0%	100.0%
Altered Splicing (EGFRvIII)	95.5%	91.3%

Performance Specifications (Durham Lab):

Variant Type	PPA	NPA
Rearrangements/Fusions (targeted)	96.8%	99.9%
Rearrangements/Fusions (untargeted)	100.0%	99.9%
Altered Splicing (MET Exon 14)	100.0%	100.0%
Altered Splicing (EGFRvIII)	100.0%	100.0%

3. xF / xF+ — Liquid Biopsy ctDNA Sequencing

What it is:

- **xF:** A 105-gene liquid biopsy circulating tumor DNA (ctDNA) sequencing panel
- **xF+:** A 523-gene liquid biopsy panel; additionally identifies variants potentially associated with clonal hematopoiesis of indeterminate potential (CH)

Sample type: Blood (no tissue required)

What it detects:

- Oncogenic drivers and resistance mutations
- Single nucleotide variants (SNVs), insertions and deletions (INDELs), copy number gains (CNGs), gene rearrangements
- MSI-H status, blood Tumor Mutational Burden (bTMB)

Resistance monitoring:

- Lung cancer: EGFR & ALK resistance mutations
- Breast cancer: ESR1 mutations
- Ovarian and breast cancer: BRCA1/2 reversion mutations
- Colorectal cancer: Anti-EGFR resistance alterations

Key clinical evidence:

- 9% of patients with actionable variants in a metastatic pan-cancer analysis had unique actionable alterations found in liquid biopsy that were not detected in solid tumor alone

Turnaround time:

- xF: typically within **7 days** of specimen retrieval
- xF+: typically within **7–9 days** of specimen retrieval

Ordering flexibility:

- May be ordered as a standalone test or in combination with xT CDx

- Longitudinal testing supported
- Automatic conversion option from xT CDx to xF/xF+ in the event of insufficient tumor tissue
- Orderable via Tempus Hub, paper requisition, or directly from your EHR

xF Performance Specifications (Chicago Lab):

Variant Class	VAF	Sensitivity	Specificity	LOD
SNVs	≥0.25%	98.5%	>99.9%	0.25%
INDELs	≥0.5%	98.5%	>99.9%	0.50%
CNGs	≥0.5%	>99.9%	96.2%	0.50%
Rearrangements	≥1%	94.4%	>99.9%	1%
MSI-H	N/A	31.3%	>99.9%	N/A
bTMB	N/A	39.0%	95.3%	N/A

4. xT Heme + xR — Hematologic Malignancies Panel

What it is: A comprehensive assessment of DNA (648 genes) and RNA whole transcriptome sequencing for hematologic malignancies, offering validated fusion detection, prognostic and/or diagnostic considerations, and therapeutic matching.

Reports include: Actionable information based on clinical and molecular findings, incorporating evidence from MSK OncoKB database and NCCN Guidelines.

Potential uses by indication:

Indication	Clinical Utility
Myelodysplastic Syndromes (MDS)	Aids in diagnosis, provides information on clonality in uncertain cases, assesses prognosis using IPSS-M
Acute Leukemias	Helps inform subclassification and prognosis, guides treatment decisions and risk stratifies patients based on molecular findings
Myeloproliferative Neoplasms (MPN)	Helps inform a definitive diagnosis and identifies certain mutations that have prognostic significance
Multiple Myeloma	May help inform risk categorization that may be of prognostic and/or therapeutic value

Turnaround time: 9 days from specimen receipt

Ordering flexibility:

- xT Heme and xR may be ordered as a standalone test or in combination
- Orderable via Tempus Hub, paper requisition, or directly from your EHR

5. xE — Whole Exome DNA Sequencing

What it is: A 19,000+ gene whole exome DNA sequencing panel with TMB. Add-ons available: IHC analysis includes PD-L1 and MMR proteins.

6. xM — Minimal Residual Disease (MRD) & Response Monitoring

What it is: Tumor-naïve and tumor-informed assays to detect residual disease or early cancer recurrence, and to monitor IO treatment response.

- **xM:** Tumor-naïve MRD assay
- **xM (NeXT Personal® Dx):** Tumor-informed MRD and IO monitoring assay (test by Personalis)

Note: New Medicare coverage available for select indications when coverage criteria are met.

7. Hereditary Risk Assessment

Tests available (all powered by Ambry Genetics®):

- **CancerNext®:** 40-gene panel for hereditary cancer
 - **CancerNext-Expanded®:** 77-gene panel for hereditary cancer
 - **+RNAinsight®:** Optional RNA add-on
 - **BRCAplus®:** 13-gene panel for hereditary breast cancer
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Algorithmic Tests

The Tempus algorithmic test platform leverages Tempus's molecular and clinical database, CAP/CLIA lab, and clinician network to develop tests that help inform the treatment of cancer patients.

IPS — Immune Profile Score

The Tempus Immune Profile Score (IPS) is a multimodal biomarker that can be used as a prognostic indicator for adult patients with metastatic and/or stage IV pan-solid tumor disease who are already considered candidates for immune checkpoint inhibitor (ICI) based therapy.

IPS uses DNA and RNA sequencing data from xT CDx and xR to classify patients as IPS-High or IPS-Low. Patients classified as IPS-High were found to have a higher real world overall survival (rwOS) compared to those with an IPS-Low result when treated with ICI-based regimens.

- No additional tissue required; ordered as an add-on with xT CDx & xR combination OR xT & xR combination
- Both DNA and xR RNA seq are required for processing
- Validated performance demonstrated in the following subgroups: PD-L1 (positive/negative), TMB (high/low), microsatellite instability status (MSS/MSI-high)

HRD — Homologous Recombination Deficiency

Tempus HRD predicts the probability of a patient's cancer having a phenotype characterized by the inability to repair DNA breaks via the homologous recombination repair (HRR) pathway.

- Available as an add-on for patients tested with xT CDx, xT, or xR
- For ovarian and breast cancer: result based on DNA genome-wide loss of heterozygosity (GWLOH) or evidence of biallelic BRCA1 or BRCA2 loss from xT CDx

- For other cancers: HRD score based on whole transcriptome RNA expression using data from xR
- No additional tissue required
- Matched normal sample required for breast and ovarian cancer

TO — Tumor Origin

The Tempus Tumor Origin (TO) test uses tumor RNA expression results to predict the patient's most likely cancer type(s) from 68 possible cancer types.

- Intended use: cancers of unknown primary and cases where available diagnostic information does not provide a definitive diagnosis
- Covers 68 possible cancer types and subtypes
- No additional tissue required; ordered as an add-on with xR RNA seq
- xR RNA seq is required for processing

DPYD

The Tempus DPYD test identifies patients who may be at elevated risk for toxicity and serious adverse effects from 5-FU and capecitabine.

- Applicable for colorectal, breast, pancreatic, and other GI cancers where 5-FU and/or capecitabine treatments are considered
- Covers five SNVs in DPYD genes
- No additional tissue required; ordered as add-on with xT CDx & xR combination
- Matched normal sample required for processing

UGT1A1

The Tempus UGT1A1 test identifies patients at elevated risk for toxicity from irinotecan, sacituzumab govitecan, and/or belinostat.

- Identifies certain genetic polymorphisms in the UGT1A1 gene
- Positive result genotypes: 28/28, 6/6, and 6/28
- Identifies 5 variants from 3 locations on the UGT1A1 gene
- No additional tissue required; ordered as add-on with xT CDx & xR combination
- Matched normal sample required for processing

PurIST®

Tempus PurIST® classifies pancreatic ductal adenocarcinomas (PDAC) into one of two subtypes: basal-like or classical.

- Published literature indicates patients with the basal subtype have a worse prognosis and are less likely to benefit from FOLFIRINOX therapy than classical patients
- Tempus independently validated use of the PurIST® algorithm with Tempus RNA sequencing data, demonstrating subtype stratification with distinct survival outcomes and treatment benefit
- No additional tissue required; ordered as add-on with xR RNA seq
- xR RNA seq is required for processing

Workflow & Platform Solutions

Tempus Hub

Tempus Hub is a secure online portal for providers to order tests, track order status, and view comprehensive patient results in one place. Available 24/7 via desktop, mobile app, and EHR.

Key features:

- **Simple ordering:** Comprehensive Therapy Selection (CTS) enables one-click ordering of DNA, RNA, and biomarker tests, tailored to the patient's cancer type
- **Smart reporting:** Reports integrate molecular and clinical results to guide therapies based on MSK's OncoKB™ and NCCN® guidelines, as well as clinical trial opportunities. Order Summary highlights actionable biomarkers, relevant therapies and trial options across all tests placed in a single order
- **Automatic therapy updates (Tempus Refresh):** Automatically updates patient results with new biomarker-matched therapies as they become available, with no new sample required
- **Real-time order tracking:** Clear visibility into patient orders from the homepage and order tracking tab; also available on mobile
- **AI-enabled capabilities:** Tempus One is integrated directly within Hub
- **Patient timeline:** Quickly access specific moments in a patient's medical history — prior therapies, pathology reports, and radiology results — all summarized from clinical notes and records

Tempus One — AI Clinical Assistant

Tempus One is a generative AI-enabled clinical assistant that provides access to patient insights directly at the provider's fingertips. Available in Tempus Hub and directly within the EHR.

Capabilities:

- Quick access to patient order status and reports, including actionable biomarkers
- Summarized patient history, treatment journey, and biomarker status
- Rapid filtering of patient incidence by alteration, gene, or diagnoses (EHR only)
- Effortless documentation: real-time transcriptions and intelligent notes saved directly to the patient's chart (EHR only)
- Request test add-ons: confirmatory germline tests, algorithmic tests, IHC tests, or cancel existing orders
- Streamline prior authorizations and anticipate next steps
- Search for clinical trials based on specific patient enrollment criteria (EHR only)
- Quick lookup of patient medical history from clinical notes and records
- Connect with a live Tempus representative if needed

Provider testimonials (from tempus.com):

- "Asking Tempus One a verbal question about the status of a test, results for a specific patient, or to look at trends across several patients is super intuitive and easy to use. It has helped with efficiently getting the real-time NGS answers I need in the clinic without the burden of digging through the chart or individual report." — Thomas George, MD, FACP, University of Florida Cancer Center
- "Tempus One demonstrates the power of assistive technologies by simplifying the often complex process of ordering, obtaining, and interpreting genomic testing." — Sandip Patel, MD, UC San Diego Health
- "Tempus One is a great platform that allows me to personalize my patients care — in my office, in the community setting, at my fingertips — and to me, that is the future of oncology." — Kamal Sharma, DO, The Oncology Institute

EHR Integration

Easy access to current treatment options and relevant insights through Tempus's integration and connectivity capabilities. Tempus One is available directly within the EHR.

Clinical Trial Matching — Tempus TIME

Tempus TIME is a clinical research program powering smart sites and creating a network designed to support and accelerate clinical development.

Key metrics:

- 100 sites representing 1,000+ clinical locations
- 1,500+ clinical trials have been active
- 2,000+ patients consented into trials
- On average, TIME completes trial onboarding in 10 weeks or less
- Site activation in as little as a few days with Just-in-Time (JIT) model

How it works:

- Advanced trial matching services combine large language models, machine learning, and clinical insight
- Clinical team of experts pre-screens matched patients and presents potentially eligible patients with white-glove service to each site
- Generative AI scans structured and unstructured clinical notes to enhance identification of patients eligible for trial participation

Tempus TIME Precision Network: A group of investigators dedicated to supporting Phase I trials, with a focus on rapid site activation and patient enrollment at Phase I-capable sites across the US.

Real-world outcome: TriHealth achieved 64% annual increase in patients enrolled in clinical trials, with Tempus TIME driving 95% of growth.

Financial Assistance

Tempus offers a transparent financial assistance program for all U.S. patients. All U.S.-based patients are eligible to apply regardless of insurance status. For international patients, a self-pay option is available.

- Applications submitted online at access.tempus.com
 - If approved, maximum out-of-pocket cost is provided immediately
 - Contact billing@tempus.com for billing questions
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EHR Integration & Connectivity

Tempus is a leader in EHR integrations, helping partners incorporate actionable data and insights into clinical workflows to improve patient care and accelerate precision medicine programs.

Scale:

- 600+ direct data connections across 5,000+ healthcare institutions that order Tempus products and services
- First NGS laboratory to deliver structured somatic variant results directly into Epic's Genomics module

Supported EHR systems:

- Epic (including Epic Genomics Module and Aura)
- Cerner
- CureMD
- Meditech
- NextGen

- Veradigm (formerly Allscripts)
- Flatiron's OncoEMR
- iKnowMed

Key capabilities:

- **Seamless EHR ordering and result delivery:** Place orders and receive results during the course of care and at the point of care, directly in your EHR, allowing you to remain focused on patient care
- **Unlocking structured genomic data:** Discrete genomic data powers precision medicine decision making, advanced analytics, and clinical research in your EHR and ancillary decision-support systems
- **Leveraging real world data:** Take advantage of Tempus's rapidly growing multimodal dataset capabilities to structure, unify, and normalize data for clinical trial matching, precision medicine analytics, publications, and AI development

Notable milestone: Together with Ochsner Health, Tempus was the first lab to provide discrete genomic results over Epic's Order & Results Anywhere (Aura) Network.

Provider testimonial:

"The integration of Epic and Tempus is a major advance in caring for patients with cancer. Until now in most institutions across the country, cancer genomic testing is done outside of their EHR platform. Integrating Tempus with Epic brings cancer genomic testing within the normal oncology clinical workflow. This ensures genomic testing is done with the appropriate patient, testing is not missed, and errors are avoided." — Dr. Janakiraman Subramanian, Director of Thoracic Oncology and Center for Precision Oncology at Saint Luke's Cancer Institute

Tempus+ — Institutional Program Support

Participating institutions receive access to technical support, customized to meet the unique needs and goals of their cancer center.

Digital Pathology — AI-Enabled Pathology (Paige)

Tempus offers a suite of digital pathology solutions using artificial intelligence to help increase diagnostic confidence, improve workflow efficiency, and uncover deeper insights from every sample. These solutions were acquired through Tempus's acquisition of Paige.

AI-Assisted Detection Applications

Paige Prostate — Aid in detection and grading of prostate cancer on H&E-stained whole-slide images

- **Paige Prostate Detect** | CE-IVDR, FDA-AUTHORIZED: First AI-based software application in pathology to receive FDA authorization to aid in the primary diagnosis of prostate cancer. Assists in detection of even minute foci suspicious for cancer. Decreases pathologists' reading times by 20%
- **Paige Prostate Grade & Quantify** | CE-IVDR, RUO: Categorizes areas suspicious for cancer by Gleason pattern, providing predictions for primary and secondary Gleason grades, plus percentage and linear measurements of tumor burden
- **Paige Prostate Perineural Invasion (PNI)** | CE-IVDR, RUO: Aids in detection of suspicious foci around nerve fibers to identify presence of PNI

Paige Breast — AI-powered applications for research use only to support breast cancer detection and staging

- **Paige Breast Detect & Neoplasm** | RUO: Highlights areas most likely to harbor cancer and in situ neoplasms

- **Paige Breast Mitosis** | RUO: Identifies mitotic hotspots and provides count and density of mitotic instances
- **Paige Breast Lymph Node** | RUO: Detects foci suspicious for breast cancer in lymph node specimens. Use of this tool improved pathologist sensitivity while cutting slide read times by over half

Paige PanCancer — Identifies foci suspicious for cancer across more than 40 cancer types

- **Paige PanCancer Detect** | RUO: Supports case triage with prediagnostic screening of all biopsy and resected tissue, or postdiagnostic QC. Achieved 0.95 specimen-level AUC across nine common and seven rare cancers

In Limited Feedback Release (RUO):

- Paige Breast Detect & Subtype
- Paige GI Detect & Subtype
- Paige GYN Detect & Subtype
- Paige Bladder Detect & Subtype

Paige Predict — Tissue Optimization & Biomarker Prediction

Paige Predict: Tissue Optimization | RUO: Forecasts total nucleic acid (TNA) yield from an H&E image to identify samples at risk of NGS failure

Paige Predict: Biomarker Prediction | RUO: AI application predicting the status of ~1,600 biomarkers across 505 genes from a single H&E slide. Provides results in approximately 5 minutes to accelerate sample selection for testing across NGS, hotspot testing, IHC, and special stains

Deployed at Tempus:

- Tissue optimization: Integrated into Tempus's clinical laboratory workflow, reducing tissue utilization by 18% and expected to lower the overall QNS (quantity not sufficient) rate by 15%
- Biomarker prediction: Validated for clinical use as a Tempus laboratory developed test, performed for eligible xT QNS cases to help inform confirmatory testing strategies

Deployment Options

- Integration with existing digital pathology platform (scanner, IMS, and viewer)
- Paige IMS: A secure, cloud-native platform including the FullFocus® Image Viewer, FullFolio™ case management, and an application marketplace for third-party AI
- Hardware agnostic: Integrates with Philips, Leica, and Hamamatsu scanners

Provider Testimonials

- "Tempus is more of a one-stop shop. Rather than sending patients to two or three different locations in order to get the same information, we can get all of it through Tempus. It is so much more convenient to get everything done with one test, one company, one set of results – instead of mixing and matching." — Jeffrey A James, DO, Colorado GynOnc
- "I am not just looking at Tempus as our partner in treatment, I see Tempus as our partner in prevention." — James F Maher, MD, TriHealth
- "The IPS score can be very helpful in these settings because it gives us an added layer of understanding of who might have better outcomes to immunotherapy." — Frank Weinberg, MD, University of Illinois Chicago
- "In the rapidly changing world of cancer care and precision medicine, Tempus is the established leader. They provide rapid, accurate results while also continuing to develop innovative strategies to improve patient outcomes." — Joseph K Hofmeister, MD, OhioHealth

- "Breast cancer care has involved use of biomarkers for years in order to determine appropriate therapy. Genomic information has now been added to the list of crucial biomarkers needed to know how to treat metastatic breast cancer patients." — Megan Kruse, MD, Cleveland Clinic

Source: tempus.com — All content scraped directly from Tempus website pages, May 2026. No AI-generated content included.